## FitTrack Personalized Fitness Recommendations

# Introduction

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# Description

FitSense is a secure REST API for workout tracking and personalized fitness guidance. Built with Node.js, Express, MongoDB and Mongoose, it supports account authentication, workout management, search, and AI recommendations through Google Gemini.

# Scenario-Based Case Study

### Background

Rahul tracks workouts in notebooks but struggles to organize his history, measure progress, and choose exercises suited to his goals.

### Problem

### Solution

FitSense centralizes workout records and provides authenticated APIs to create, view, search, update and delete them. Gemini uses a user's age, goal and experience to generate recommendations and fitness insights.

JWT authentication, bcrypt password hashing, input validation and centralized error handling protect user data.

# Usage

## User Usage

Users register and sign in to receive a JWT, then use protected endpoints to manage workouts and request AI guidance.

Using authenticated APIs, users can:

- Request personalized workout recommendations and fitness insights.
- Obtain AI-generated fitness insights based on workout statistics.

## AI Usage

The Gemini service uses age, fitness goal and experience level to create a workout plan, weekly suggestions, safety tips and motivation. It can also summarize workout totals, duration and calories.

Users provide information such as:

Gemini processes this information and returns:

Similarly, users can submit workout statistics including:

The AI module analyzes these values and produces personalized fitness insights.

# Outcome

FitSense replaces manual workout logs with secure tracking and AI-supported recommendations. Its modular design can later support nutrition, wearable devices, reminders and analytics.

## 1. Software Requirements

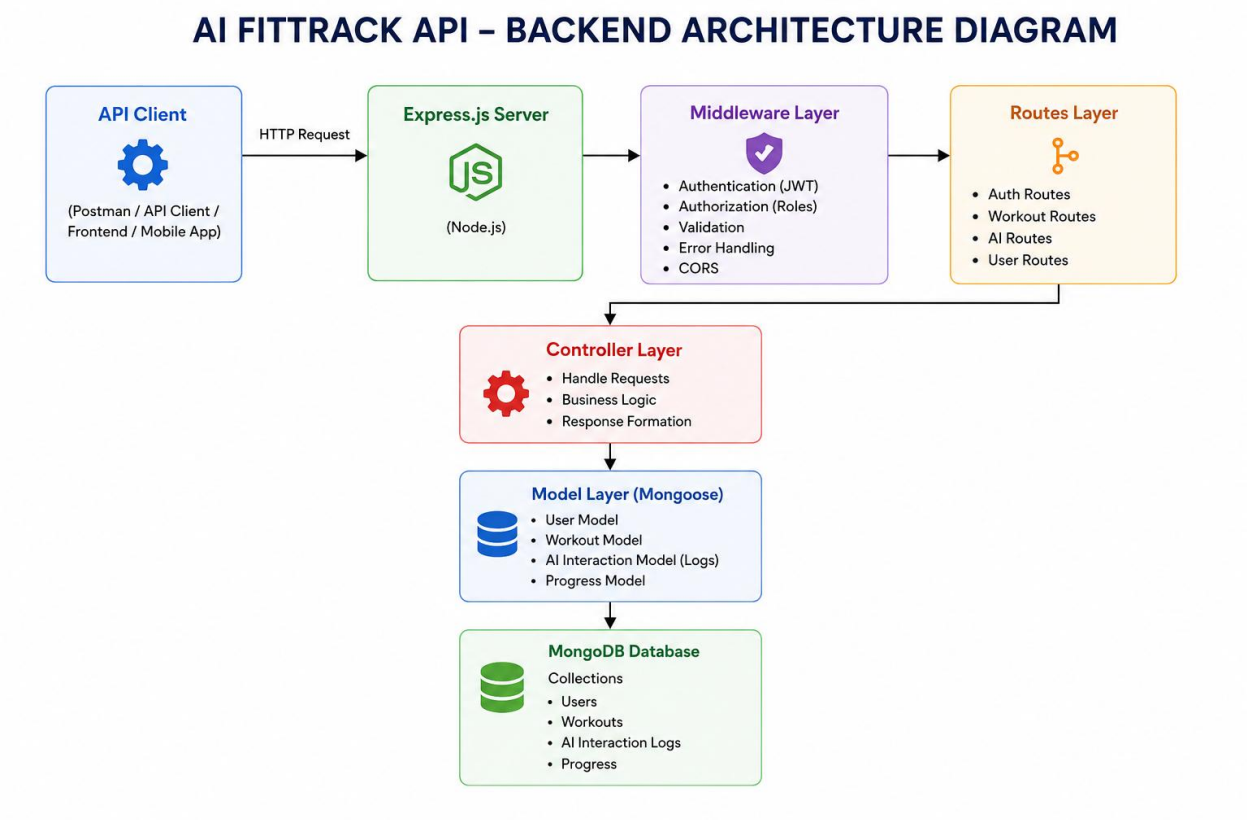
Windows, macOS or Linux; Node.js 16+, npm 8+, Express.js, MongoDB, Postman or Thunder Client, and a code editor.

- **Node.js (v16 or above):** Runtime ecosystem running server logic and routing infrastructure.
- **npm (v8 or above):** Node package manager to control operational dependencies.
- **Express.js:** Lightweight routing web framework to construct backend RESTful entry points.
- **MongoDB:** Document NoSQL engine storing users, posts, categories, comments, and analytics metrics.
- **Postman:** API verification toolkit to assert schema validation responses across administrative route shields.
- **Code Editor:** Visual Studio Code or similar IDE environment.

## 2. Hardware Requirements

- Intel Core i5 (8th generation) / AMD Ryzen 5 or equivalent; 8 GB RAM minimum (16 GB recommended); 1 GB available storage.
- **RAM:** 8 GB minimum (16 GB recommended for concurrent instances of MongoDB, processing layers, and execution nodes).
- **Storage:** 1 GB of available disk workspace.

# Epic 1: Project Architecture




FitSense uses an MVC-style backend. Requests pass from the client through Express routes and JWT middleware to controllers, services, MongoDB models or the Gemini service. Responses return as JSON.

The route, controller, service and data layers separate responsibilities and support maintenance and extension.

This layered approach improves modularity, security, and maintainability while ensuring that each component has a single responsibility.

## Overall Architecture Components

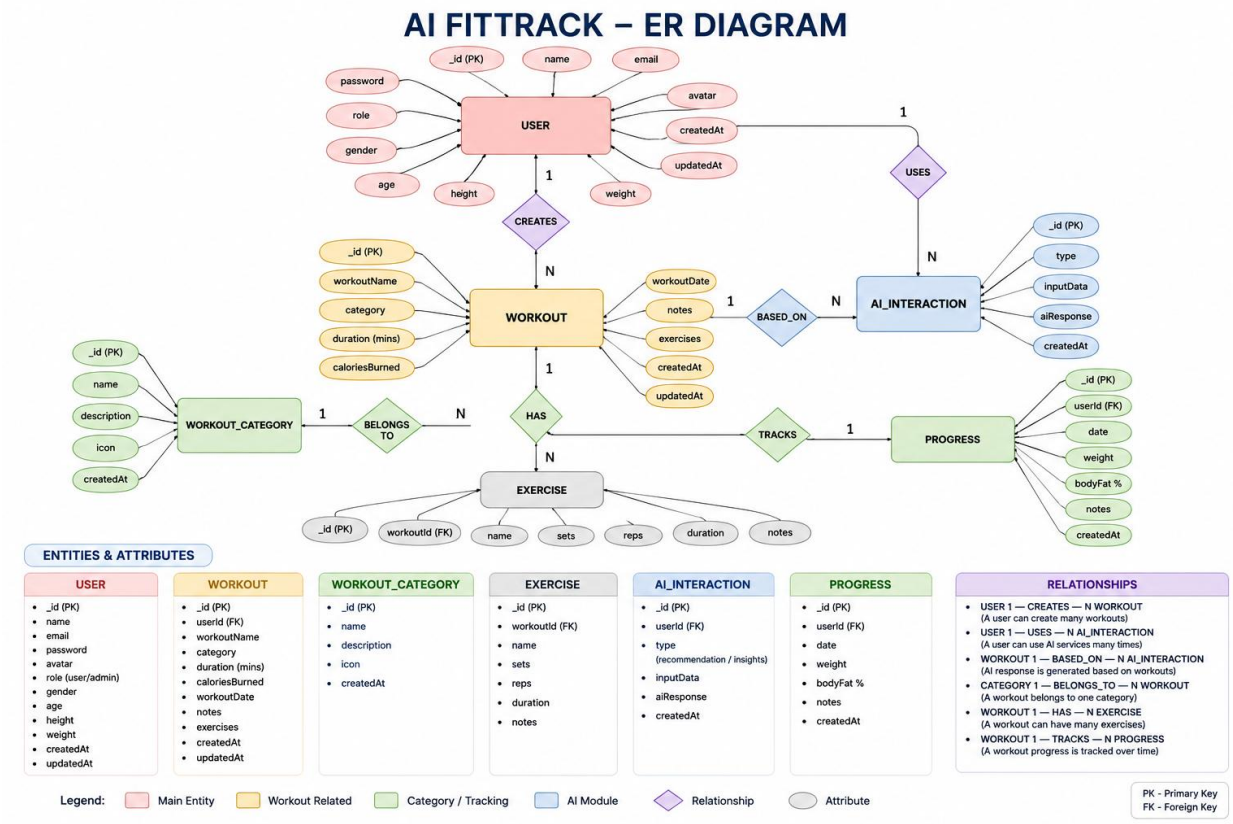
Each component communicates through well-defined interfaces, ensuring loose coupling and easier future enhancements.

# Components Definition

The backend separates client access, routing, authentication, controllers, services and data storage. Each layer handles a focused part of a request.

# The data model has two core entities: User and Workout. One user can own many workouts; each workout belongs to one user.

The AI FitTrack backend consists of two primary collections that are logically connected through a one-to-many relationship.




## 1. User Entity

Stores account and profile information, including a unique email and hashed password.

| Field | Type | Description |
|---|---|---|
| _id | ObjectId | Primary Key |
| name | String | User Name |
| email | String | Unique Email Address |
| password | String | Encrypted Password |
| createdAt | Date | Record Creation Time |
| updatedAt | Date | Last Updated Time |

## 2. Workout Entity

Stores workout details and a reference to the owning user.

| Field | Type | Description |
|---|---|---|
| _id | ObjectId | Primary Key |
| user | ObjectId | References User Collection |
| workoutName | String | Workout Name |
| category | String | Workout Category |
| duration | Number | Duration (Minutes) |
| caloriesBurned | Number | Calories Burned |
| workoutDate | Date | Workout Date |

| | | |
|---|---|---|
| createdAt | Date | Created Timestamp |
| updatedAt | Date | Updated Timestamp |

**Relationship Type:** One-to-Many

One user can create multiple workout records; each record belongs to one authenticated user.

This relationship is maintained using the **user** field inside the Workout schema, which stores the ObjectId of the corresponding user.

# Key Features

Core capabilities include secure authentication, workout CRUD and search, Gemini recommendations and insights, RESTful JSON APIs, and centralized error handling.

### 1. Secure Authentication

### 2. Workout Management

Each workout is linked to its authenticated user.

### 4. Google Gemini Integration

The backend communicates with Google Gemini AI to generate intelligent workout recommendations.

### 5. AI Fitness Insights

The AI module analyzes workout statistics.

### 6. Centralized Error Handling

### 7. RESTful API Design

### 8. Modular Architecture

# Roles and Responsibilities

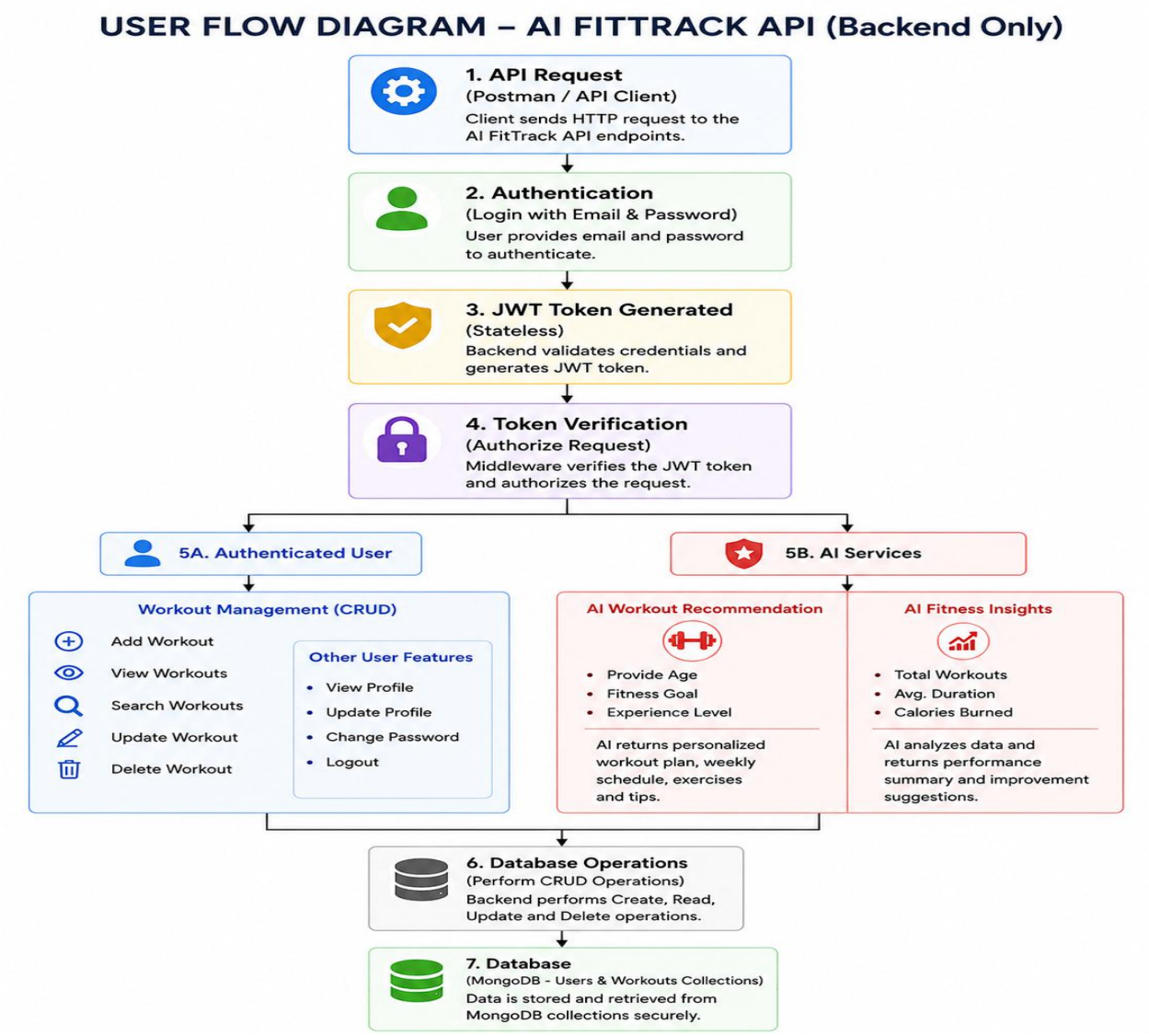
The AI FitTrack backend supports one primary role.

### `Authenticated User`

### Permissions

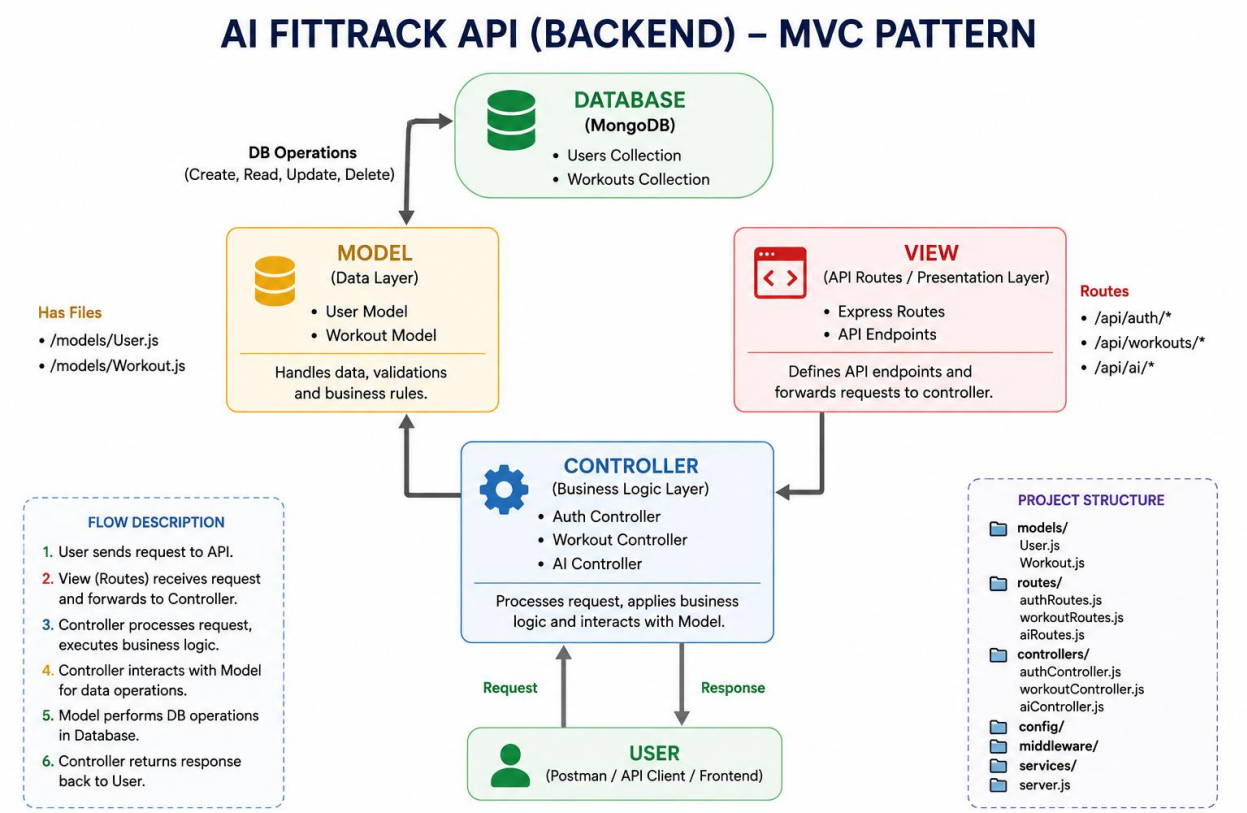
| Operation | Permission |
|---|---|
| Register | Yes |
| Login | Yes |
| View Profile | Yes |
| Add Workout | Yes |
| Get Workouts | Yes |
| Search Workouts | Yes |
| Update Workout | Yes |
| Delete Workout | Yes |
| AI Recommendation | Yes |
| Fitness Insights | Yes |

# User Flow




# MVC Architecture Pattern

The FitTrack AI  is designed using the **Model–View–Controller (MVC)** architecture pattern. This architecture separates application logic into independent components, improving code organization, maintainability, and scalability.




## Model Layer (Data Layer)

The Model Layer defines the structure of the application's data using Mongoose schemas. It communicates directly with MongoDB and performs database operations.

## View Layer (API Interface)

## Controller Layer (Business Logic)

The Controller Layer acts as the intermediary between the View and Model. It processes incoming requests, validates input, invokes services and models, and sends structured JSON responses back to the client.

The separation of these three layers ensures loose coupling, simplifies testing, enhances code readability, and makes the application easier to extend with future features.

# Epic 2: Project Execution

· Install ThunderClient (VS Code extension).

· Start the backend server:

## “npm start”

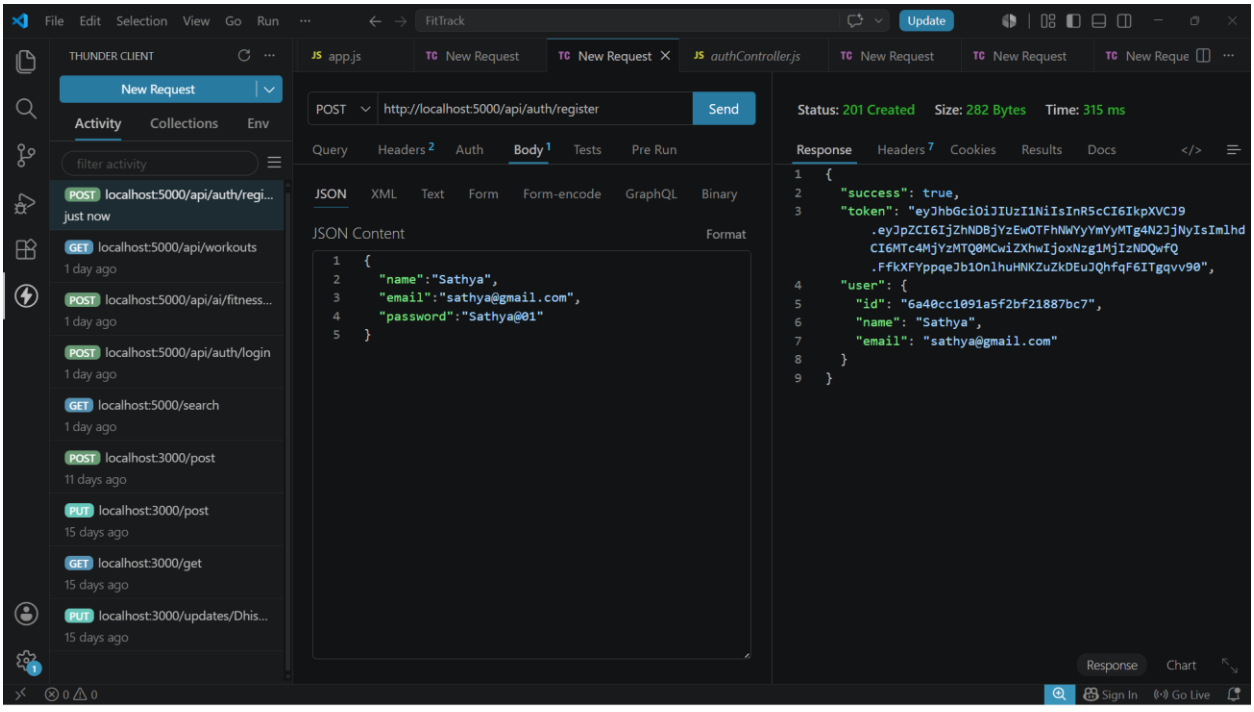
### API Testing

Use Postman or Thunder Client to verify authentication, workout management and AI endpoints.

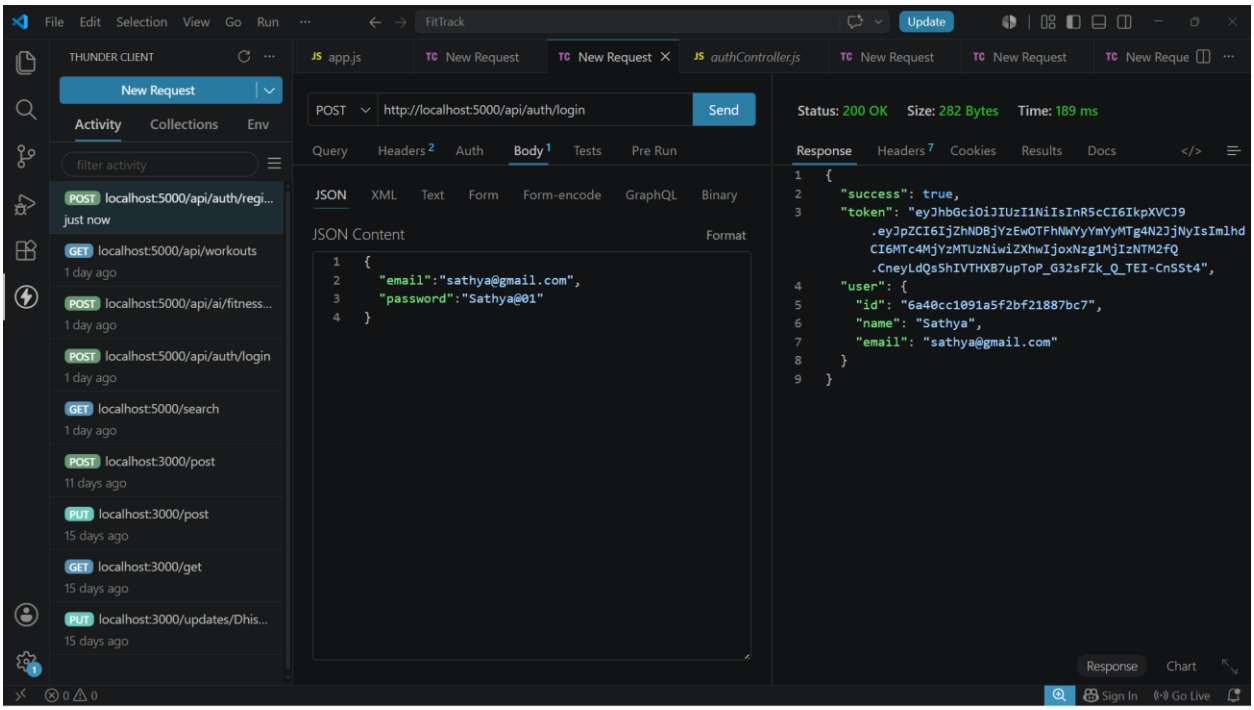
Testing is performed in the following order:

# Authentication API Testing

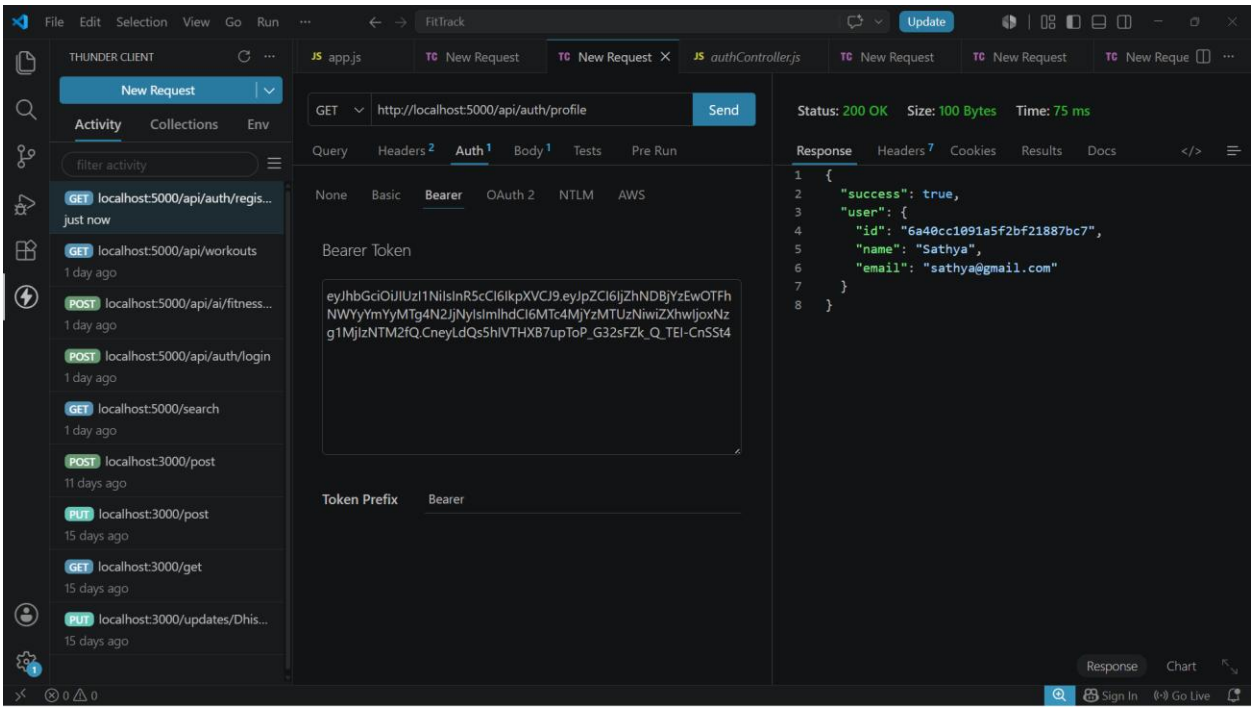
### Register User



## Login User

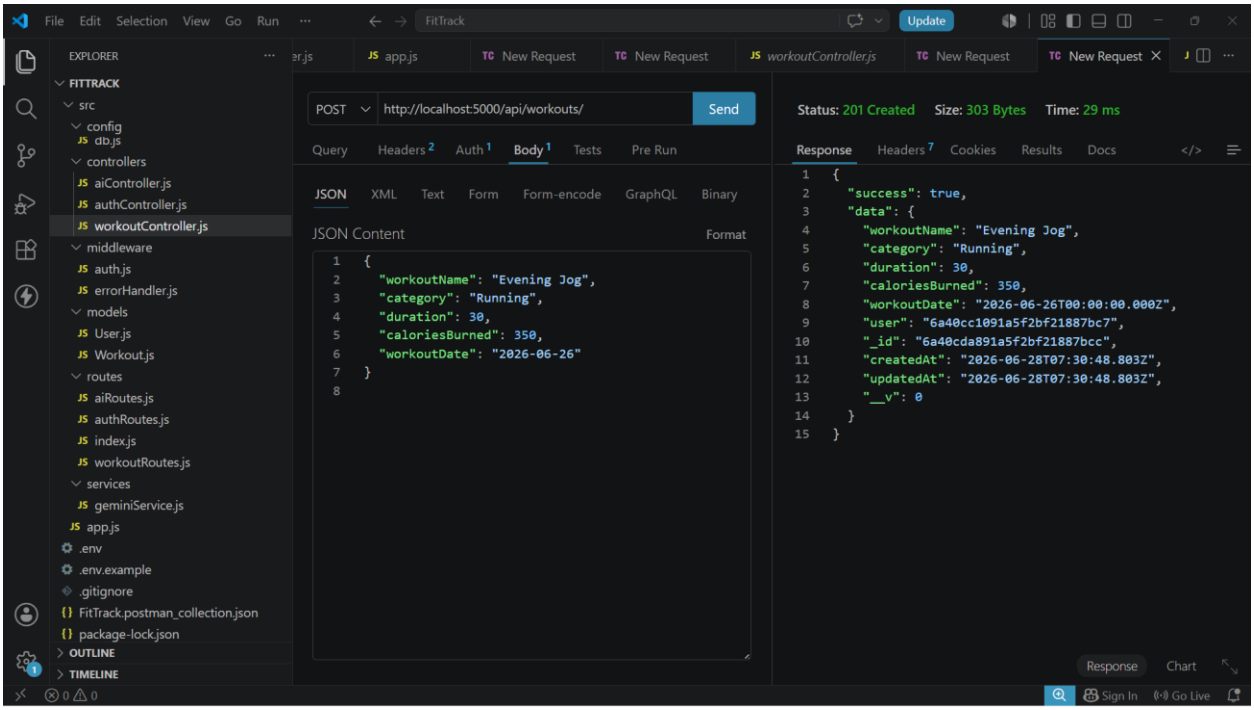



## Get Profile

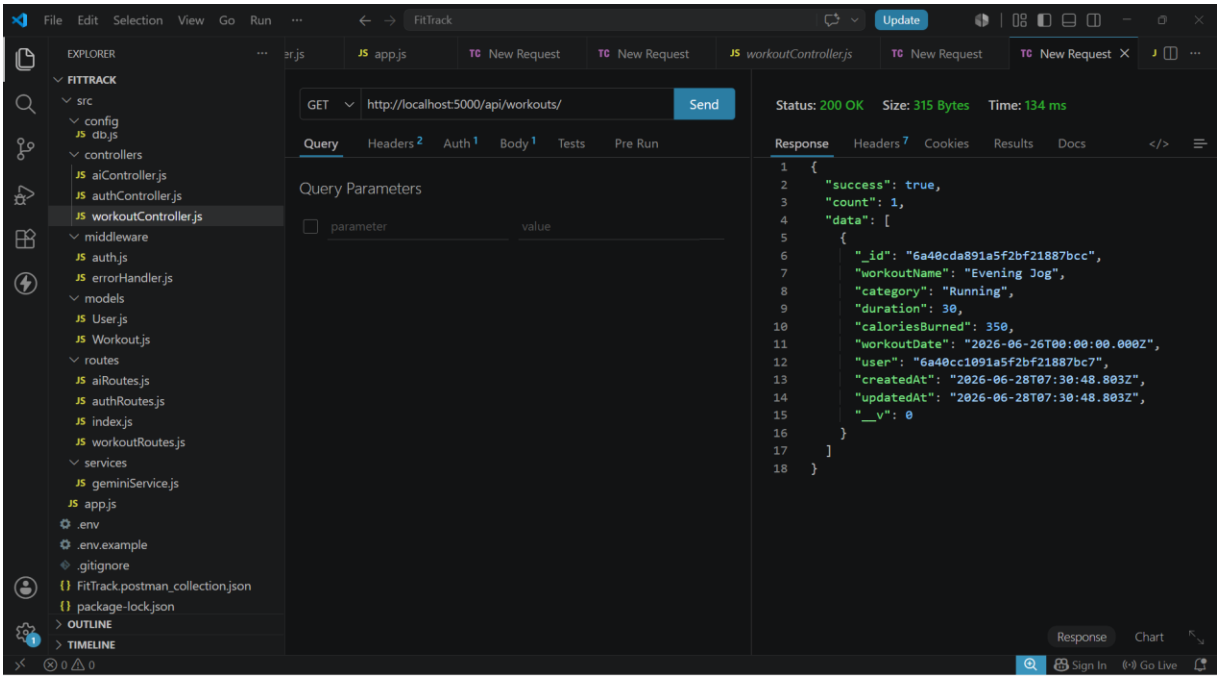


# Workout API Testing

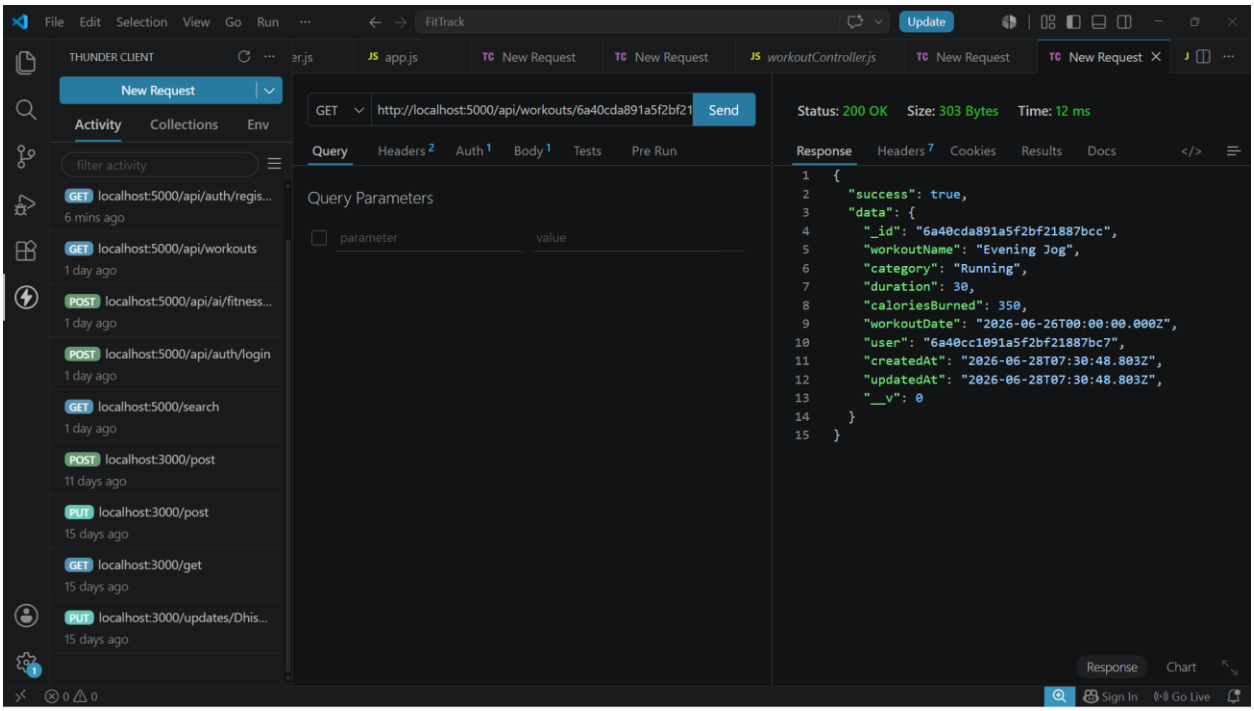
## Add Workout



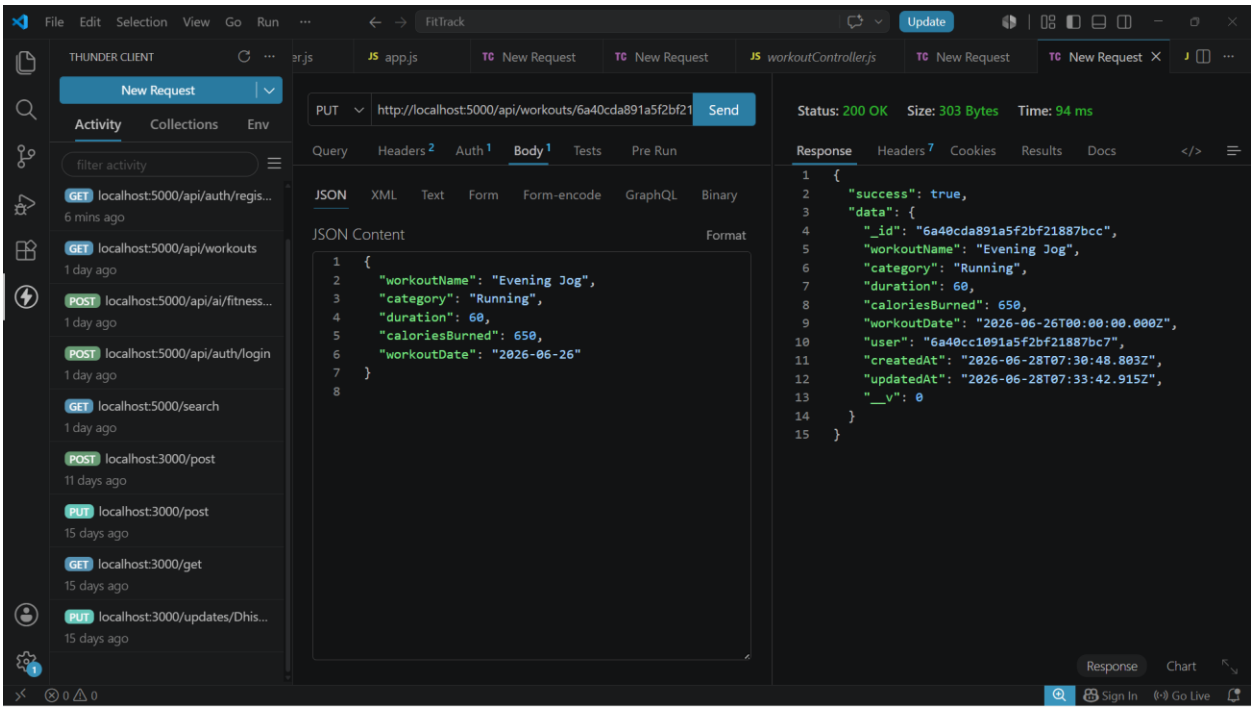

## Get All Workouts



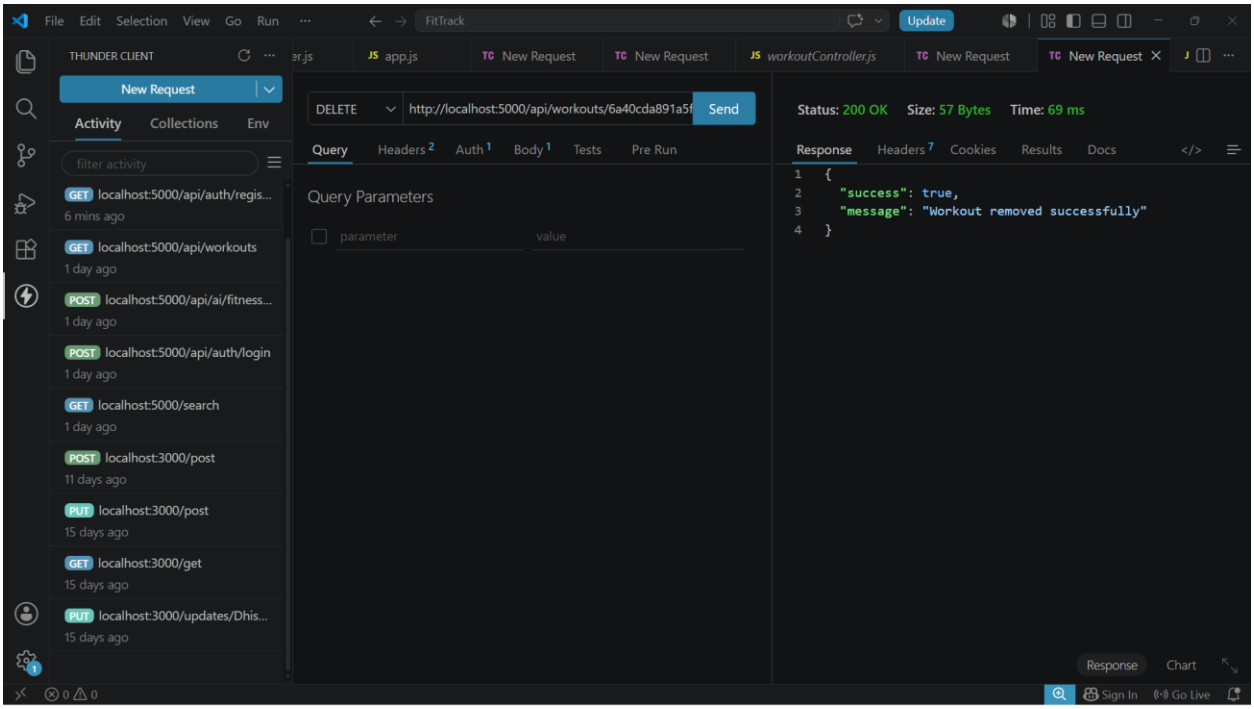
## Get Workout By ID




## Update Workout

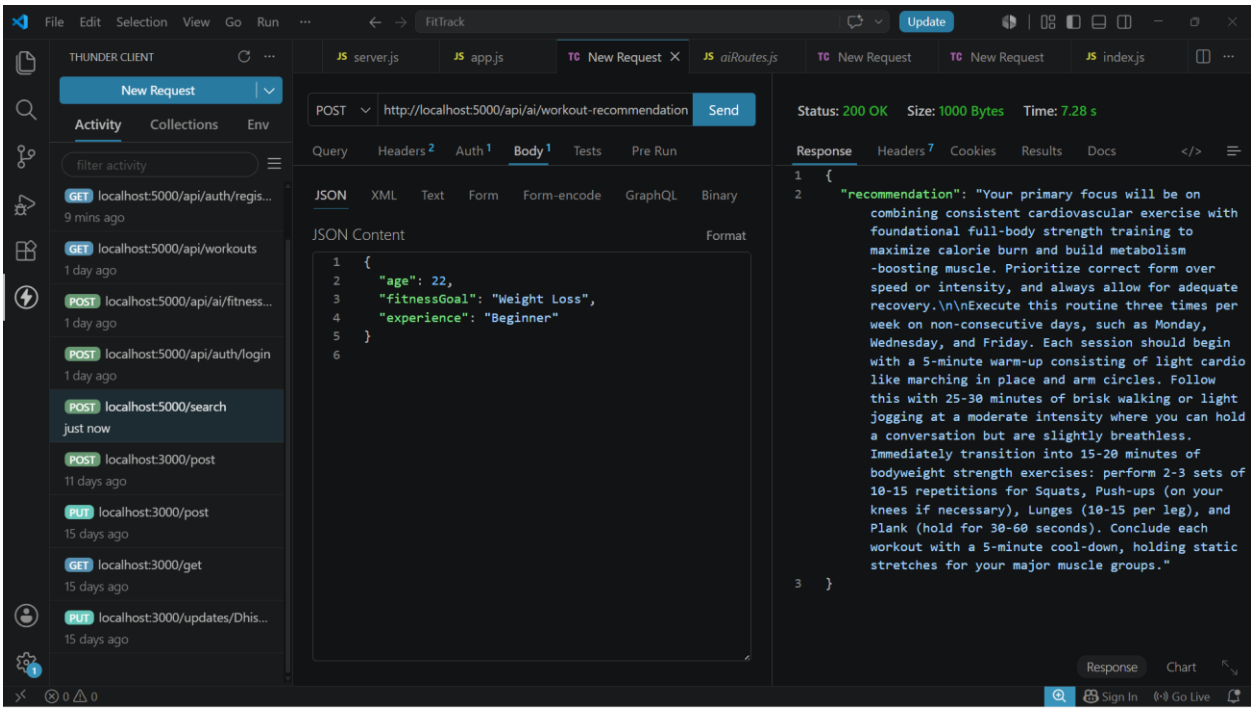


### Delete Workout

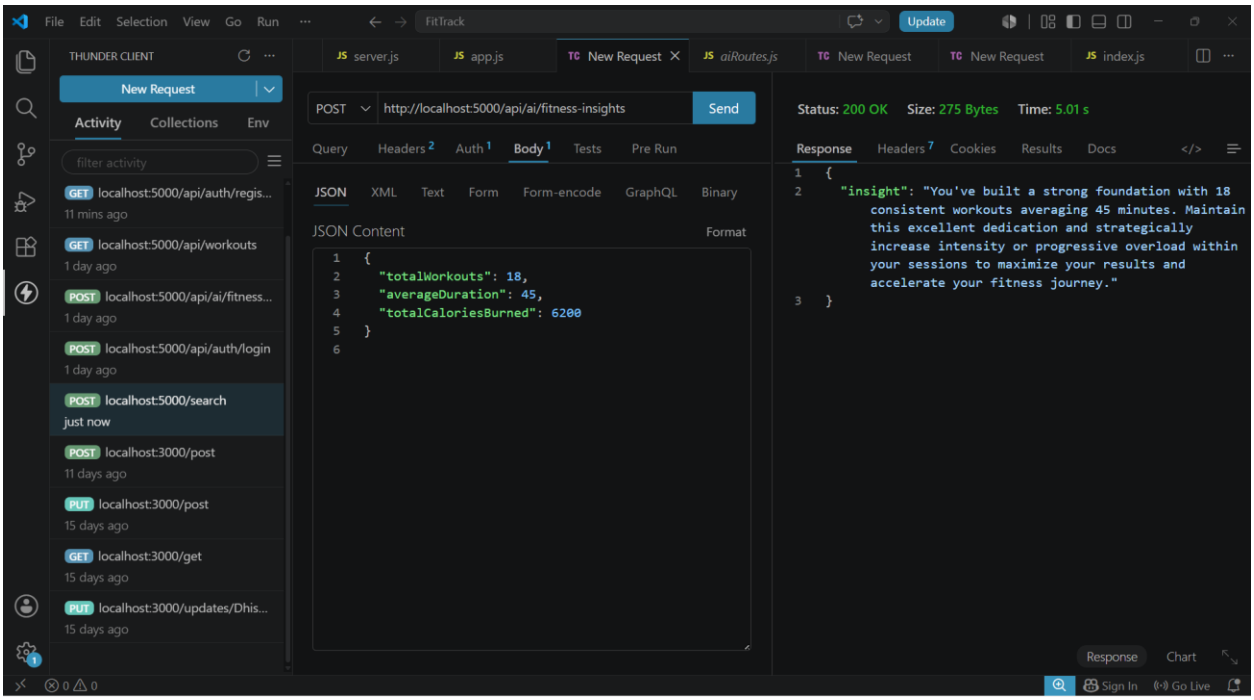



# AI API Testing

### Workout Recommendation



**Fitness Insights**




## Demo video link:

https://drive.google.com/drive/folders/1vmpVoIapOuIW4wGY-P1bnIV_OJqLNxVY?usp=drive_link

## Code files:

https://drive.google.com/drive/folders/1YDq2Uluv2RAj3nMnvFVuVX_OpPmmCWtV?usp=drive_link

## Code repository link:

https://github.com/tanish6106/AI-FitTrack.git